

Supplemental Material:

Auditing Trajectory-Based Transfer Diagnostics in Variational Quantum Learning

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This Supplemental Material collects the supporting analyses referenced in the main text: the concentration of the identity prefactor (Sec. S1), the one-step versus multi-step identity check (Sec. S2), the association of alignment with the query loss rather than the gap (Sec. S3), the near-zero-mean-transfer character of the task family (Sec. S4), and four robustness checks—a term-type-stratified split, additional Hamiltonian families and entangler topologies, a finite-shot variance proxy, and a random initialization (Sec. S5); and a synthetic implementation check showing that the joint adjustment does not mechanically zero an explicitly injected direct A component (Sec. S6). All quantities use the definitions, estimators, and hierarchical-bootstrap procedure of the main text. Rank correlations are Spearman coefficients; joint controls are partial rank correlations $\rho(A, G | I_{\text{val}}, L_{\text{val}}(\theta_0))$ unless stated otherwise. The underlying per-task tables are provided in the accompanying data repository.

S1. CONCENTRATION OF THE IDENTITY PREFACTOR

The first-order identity of the main text reads $I_{\text{val}} \approx \alpha A q$ with prefactor $q = \|g_{\text{tr}}\| \|g_{\text{val}}\|$. The small-relative-variation limit of q is a statement about the relative spread of q , not of $\|g_{\text{tr}}\|$ alone. Table SI reports the pooled coefficient of variation of $\|g_{\text{tr}}\|$, $\|g_{\text{val}}\|$, and q across tasks at each system size. Both gradient norms concentrate with scale, and the product q —being the product of two fluctuating norms—has a larger but likewise decreasing coefficient of variation. This is the regime in which the ranking of I_{val} approaches, in the small-relative-variation limit, that of A .

TABLE SI. Pooled coefficient of variation of the two gradient norms and of the identity prefactor $q = \|g_{\text{tr}}\| \|g_{\text{val}}\|$, as a function of system size n (pooled over depth L and seeds).

n	$\text{CV}(\ g_{\text{tr}}\)$	$\text{CV}(\ g_{\text{val}}\)$	$\text{CV}(q)$
4	0.336	0.361	0.551
6	0.285	0.282	0.466
8	0.246	0.261	0.406
10	0.264	0.261	0.451

S2. ONE-STEP VERSUS MULTI-STEP IDENTITY

The identity is a first-order (tangent) expression; it is at its most accurate in the one-step limit. Table SII compares the rank correlation between the alignment A and the validation improvement measured after a single inner step ($I_{\text{val}}^{(1)}$) with that measured after the full $K = 10$ inner steps ($I_{\text{val}}^{(K)}$). At $K = 1$ the correlation is near-perfect and essentially flat in n (0.95–0.97), confirming that the tangent expression captures the one-step improvement directly. At $K = 10$ the correlation is lower and rises with scale (0.60 to 0.85): as gradient magnitudes and adaptation steps shrink while norm variability also changes, the full trajectory moves closer to its initial tangent, and the identity increasingly predicts the multi-step improvement as well. The tightening reported in the main text is therefore the convergence of the K -step improvement toward the one-step tangent, not a change in the identity itself.

S3. ALIGNMENT VERSUS THE QUERY LOSS

The main text controls the apparent association between A and the observable-generalization gap G . The same conclusion holds when the target is the query loss $L_{\text{val}}(\theta_K)$ itself rather than the gap. Table SIII reports $\rho(A, L_{\text{val}}(\theta_K))$ raw, controlling for the local improvement I_{val} , and controlling jointly for I_{val} and the initial baseline $L_{\text{val}}(\theta_0)$. The

TABLE SII. Rank correlation of the support-validation alignment A with the validation improvement after one inner step ($I_{\text{val}}^{(1)}$) and after $K = 10$ inner steps ($I_{\text{val}}^{(K)}$), pooled over depth and seeds.

n	$\rho(A, I_{\text{val}}^{(1)})$	$\rho(A, I_{\text{val}}^{(K)})$
4	+0.95	+0.60
6	+0.96	+0.68
8	+0.97	+0.76
10	+0.97	+0.85

raw association is moderate and negative but is near zero under the joint control for each pooled n ($|\rho| \leq 0.10$), so the vanishing residual is not special to the signed gap.

TABLE SIII. Alignment versus the query loss $L_{\text{val}}(\theta_K)$: raw rank correlation, controlling for the local improvement I_{val} , and the joint control additionally removing the initial baseline $L_{\text{val}}(\theta_0)$.

n	$\rho(A, L_{\text{val}}^K)$	$\rho(A, L_{\text{val}}^K I_{\text{val}})$	$\rho(A, L_{\text{val}}^K I_{\text{val}}, L_{\text{val}}^0)$
4	-0.56	-0.34	-0.06
6	-0.49	-0.39	-0.03
8	-0.45	-0.38	-0.10
10	-0.36	-0.38	-0.02

We also report the baseline-corrected gap $\Delta G = [L_{\text{val}}(\theta_K) - L_{\text{tr}}(\theta_K)] - [L_{\text{val}}(\theta_0) - L_{\text{tr}}(\theta_0)] = I_{\text{tr}} - I_{\text{val}}$, which removes the initial baseline at the level of the target. Its raw association with A is $\rho(A, \Delta G) = -0.48, -0.55, -0.53, -0.59$ at $n = 4, 6, 8, 10$; controlling for I_{val} alone removes it, $\rho(A, \Delta G | I_{\text{val}}) = +0.03, -0.03, -0.07, -0.10$.

S4. NEAR-ZERO-MEAN-TRANSFER REGIME

Table SIV quantifies the transfer content of the primary task family. Support adaptation improves the support loss substantially (mean I_{tr} from +0.41 to +0.16), but the mean validation improvement is statistically indistinguishable from zero at every size, with roughly half of tasks improving. The per-task spread of I_{val} remains nonzero and is the quantity the first-order identity predicts. This regime is empirically a near-zero-mean-transfer setting under the specified construction: coefficients are drawn independently and the linear observable loss over disjoint Pauli support and query subsets has no shared minimizer, so there is little mean observable transfer to detect.

TABLE SIV. Transfer content of the primary task family: mean and standard deviation of the validation improvement I_{val} , fraction of tasks with $I_{\text{val}} > 0$, and mean support improvement I_{tr} .

n	$\langle I_{\text{val}} \rangle$	$\text{std}(I_{\text{val}})$	$\text{frac}(I_{\text{val}} > 0)$	$\langle I_{\text{tr}} \rangle$
4	-0.008	0.283	0.47	0.410
6	+0.002	0.161	0.51	0.313
8	-0.003	0.096	0.49	0.242
10	+0.001	0.060	0.50	0.158

S5. ROBUSTNESS CHECKS

The four checks below vary one ingredient of the setup at a time and confirm that the identity link and the vanishing joint-controlled gap association both persist.

A. Term-type-stratified support/query split

To rule out that the gap is driven by an unbalanced split, we stratify the support/query split so that XX , YY , ZZ , and Z terms are balanced across the two subsets, and compare against the uniform (random) split. Table SV shows that both splits preserve the high identity link and the near-zero joint-controlled gap association.

TABLE SV. Uniform versus term-type-stratified support/query split. The identity link $\rho(A, I_{\text{val}})$ and the joint-controlled gap association $\rho(A, G | I_{\text{val}}, L_{\text{val}}(\theta_0))$ are preserved under stratification.

n	split	$\rho(A, I_{\text{val}})$	$\rho(A, G I_{\text{val}}, L_{\text{val}}^0)$
4	uniform	0.35	−0.29
4	stratified	0.50	+0.10
6	uniform	0.81	−0.03
6	stratified	0.72	−0.05
8	uniform	0.76	+0.04
8	stratified	0.86	−0.01

B. Additional Hamiltonian families and topologies

We repeat the analysis at $n = 6$ for three Hamiltonian families—the anisotropic XYZ family of the main text, a transverse-field Ising model (TFIM), and a fixed random-local-Pauli family—each with a linear-chain and a ring entangler. Table SVI shows that the identity link stays high ($\rho(A, I_{\text{val}}) = 0.49$ – 0.91) and the joint-controlled gap association stays near zero (-0.17 to $+0.13$) throughout, so the observed identity-link and adjusted-null association pattern is not specific to the XYZ family or the linear chain.

TABLE SVI. Identity link and joint-controlled gap association across three Hamiltonian families and two entangler topologies ($n = 6$, $L = 3$).

family	topology	$\rho(A, I_{\text{val}})$	$\rho(A, G I_{\text{val}}, L_{\text{val}}^0)$
XYZ	linear	0.81	−0.03
XYZ	ring	0.89	+0.08
TFIM	linear	0.49	+0.13
TFIM	ring	0.77	+0.07
rand. Pauli	linear	0.78	−0.03
rand. Pauli	ring	0.91	−0.17

C. Finite-shot variance proxy

The observable-variance proxy V_{tr} is defined with a finite shot budget. To confirm that finite sampling does not remove the effect, we compare V_{tr} estimated at 100 and 1000 shots against the exact ($N_{\text{shot}} \rightarrow \infty$) value across tasks. The rank agreement is $\rho = +0.84$ at 100 shots and $\rho = +0.98$ at 1000 shots, with a small positive sampling bias (mean offset $+0.003$ and $+0.001$, respectively). Finite-shot statistics therefore add, rather than remove, sources of the apparent association.

D. Random initialization

Finally, we replace the Reptile meta-initialization with a plain random initialization ($\theta \sim \mathcal{N}(0, 0.1^2)$, no meta-training) at $n = 6$, $L = 3$. The identity link is $\rho(A, I_{\text{val}}) = +0.74$, the raw gap association is $\rho(A, G) = -0.27$, and it collapses under the joint control to $+0.07$ —matching the Reptile arm ($0.68, -0.42, -0.01$). The identity and

the adjusted-null pattern are thus consistent with the first-order variational geometry rather than specific to the meta-initialization.

S6. SYNTHETIC IMPLEMENTATION CHECK: INJECTED COMPONENTS ARE NOT MECHANICALLY ZEROED

Every demonstration in the main text is a collapse case, so one may ask whether the joint control is simply too aggressive—whether it would erase an explicitly injected alignment–gap component. A closed-form synthetic ensemble answers this. Per task we draw an alignment A , a norm product q with CV ≈ 0.45 (matching the experiments), the identity-channel improvement $I_{\text{val}} = \alpha Aq + \varepsilon$, and an independent baseline $L_{\text{val}}(\theta_0)$; the gap is then constructed in two ways. In the *channels-only* construction, $G = -0.8 I_{\text{val}} + 0.8 L_{\text{val}}(\theta_0) + \varepsilon'$, so any A – G association flows entirely through the two controlled channels. In the *direct-channel* construction a direct term $-0.5 A$ is added, representing a real alignment→gap effect beyond local improvement and baseline. Three seeds of 150 tasks each give Table SVII: the channels-only association collapses under the joint control ($-0.65 \rightarrow -0.06$), exactly as in the physical experiments, while the direct-channel association survives it ($-0.90 \rightarrow -0.62$; per-seed range $[-0.66, -0.58]$). The numerical values depend on the injected channel strength: sweeping the direct coefficient from 0.1 to 0.8 (channel noise s.d. 0.10) gives surviving joint-controlled correlations of -0.21 , -0.33 , -0.45 , -0.62 , and -0.77 , so survival is robust down to injected strengths comparable to the channel noise. The controls of the audit protocol therefore remove channel-mediated associations while not mechanically forcing an explicitly injected component to zero: a diagnostic that survives them remains a candidate requiring independent validation.

TABLE SVII. Synthetic implementation check. Rank correlation of alignment with the gap, raw and under the joint control, when the gap contains only the local-improvement and baseline channels versus when an explicitly injected direct alignment channel is added (3 seeds $\times$ 150 tasks, pooled).

construction	$\rho(A, G)$	$\rho(A, G I_{\text{val}})$	$\rho(A, G I_{\text{val}}, L_{\text{val}}^0)$
channels only	-0.65	-0.01	-0.06
direct channel	-0.90	-0.45	-0.62